

Data Mining

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UNIT 1: INTRODUCTION TO DATA MINING

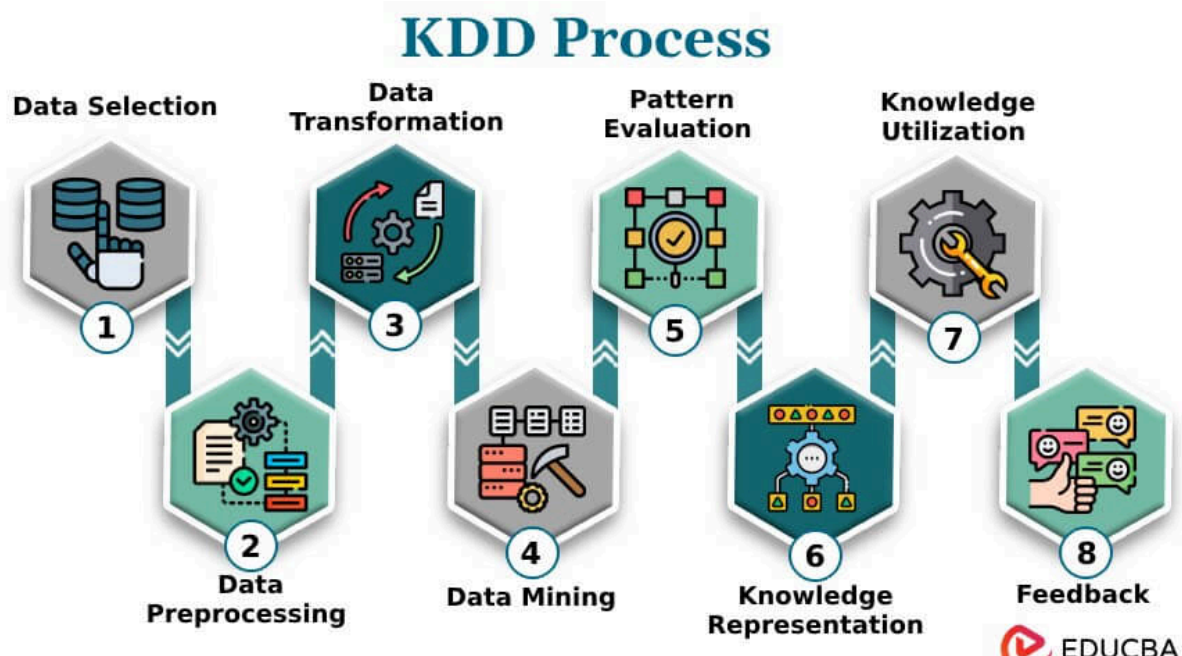
KDD - Knowledge Discovery in Databases

Knowledge Discovery in Databases (KDD) refers to the complete process of uncovering valuable knowledge from large datasets. It starts with the selection of relevant data, followed by preprocessing to clean and organize it, transformation to prepare it for analysis, data mining to uncover patterns and relationships, and concludes with the evaluation and interpretation of results, ultimately producing valuable knowledge or insights. KDD is widely utilized in fields like machine learning, pattern recognition, statistics, artificial intelligence, and data visualization.

KDD (Knowledge Discovery in Databases) = Entire process of extracting **useful, valid, novel, and understandable knowledge** from data

Data Mining = only one step inside KDD

The KDD process is iterative, involving repeated refinements to ensure the accuracy and reliability of the knowledge extracted.



FLOW

Data Cleaning → Data Integration → Data Selection → Data Transformation

→ Data Mining → Pattern Evaluation → Knowledge Presentation

1. DATA CLEANING

✓ Purpose:

Remove **noise, missing, inconsistent data**

♦ Sub-steps:

(i) Missing value handling

Reasons Behind Missing Values in the Dataset

Data can be missing from a dataset for several reasons and understanding the cause is important for selecting the most effective way to handle it.

Common reasons for missing data include:

- **Technical issues:** Failed data collection or errors during data transmission.
- **Human errors:** Mistakes like incorrect data entry or oversights during data processing.
- **Privacy concerns:** Missing sensitive or personal information due to confidentiality policies.
- **Data processing issues:** Errors that occur during data preparation.

Types of Missing Values

- **MCAR (Missing Completely at Random):** Missingness occurs randomly and is not related to any variable in the dataset.
- **MAR (Missing at Random):** Missingness depends on other observed variables, not on the missing value itself.
- **MNAR (Missing Not at Random):** Missingness is directly related to the value that is missing (e.g., high-income individuals not reporting income).

Methods for Identifying Missing Data

Detecting and managing missing data is important for data analysis. Let's see some useful functions for detecting, removing and replacing null values in Pandas DataFrame.

Functions	Descriptions
<code>.isnull()</code>	Detect missing values
<code>.notnull()</code>	Detect non-missing values
<code>.info()</code>	Summary with missing counts
<code>.isna()</code>	Same as <code>isnull()</code> .
<code>dropna()</code>	Remove missing rows/columns
<code>fillna()</code>	Fill missing values
<code>replace()</code>	Replace specific values
<code>drop_duplicates()</code>	Remove duplicate rows

unique()	Get unique values in a Series/DataFrame
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Representation of Missing Values in Datasets

Missing values can appear in different forms, so using a consistent and well-documented format is important for clean [data handling](#).

Common representations include:

1. **Blank cells:** Empty entries in spreadsheets or CSV files.
2. **Special values:** Placeholders like "NA", "NaN", "NULL" or numbers such as -999.
3. **Codes or flags:** Labels like "MISSING" or "UNKNOWN" to indicate specific types of missing data.

Strategies for Handling Missing Values in Data Analysis

1. Remove rows with missing values - df.dropna()

- **Column Deletion:** Removes entire columns with high percentages of missing data (>80%), as imputation may introduce too much bias.

2. Imputation - Imputation replaces missing values with estimated values to preserve the dataset's size. However, the filled values may not always represent the true data.

1. Mean, Median and Mode Imputation: This method involves replacing missing values with the mean, median or mode of the relevant variable. It's a simple approach but it doesn't account for the relationships between variables.

```
mean_imputation = df['Marks'].fillna(df['Marks'].mean())
median_imputation = df['Marks'].fillna(df['Marks'].median())
mode_imputation = df['Marks'].fillna(df['Marks'].mode().iloc[0])
```

- **.iloc[0]:** Accesses the first element of the Series which represents the mode.

2. Forward and Backward Fill : These methods fill missing values using the closest available values in the column. They are useful for ordered or time-series data.

```
forward_fill = df['Marks'].fillna(method='ffill')
backward_fill = df['Marks'].fillna(method='bfill')
```

Pros: preserves order and patterns in data

Cons: may be inaccurate when gaps are large or values differ significantly

Note:

- *Forward fill uses the last valid observation to fill missing values.*
- *Backward fill uses the next valid observation to fill missing values.*

3. Interpolation Techniques

Interpolation estimates missing values by using the trend or pattern of surrounding data points. It is more informative than simple imputation.

```
linear_interpolation = df['Marks'].interpolate(method='linear')
quadratic_interpolation = df['Marks'].interpolate(method='quadratic')
```

- `df['Marks'].interpolate(method='linear')`: This method performs linear interpolation on the 'Marks' column of the DataFrame (df).
- `df['Marks'].interpolate(method='quadratic')`: This method performs [quadratic interpolation](#) on the 'Marks' column.

Pros: captures trends, preserves relationships

Cons: assumes a pattern (linear, quadratic), may be complex

Note:

- *Linear interpolation assumes a straight line between two adjacent non-missing values.*
- *Quadratic interpolation assumes a quadratic curve that passes through three adjacent non-missing values.*

- **Constant Value:** Fills missing entries with a specific constant, such as 0 or "Unknown".
- **Advanced Imputation:**
 - **KNN Imputation:** Uses K-Nearest Neighbors to predict missing values based on similarity to other rows.
 - **Regression Imputation:** Uses linear or other regressions to estimate missing values based on relationships with other features. [GeeksforGeeks +5](#)

When to Use Which Method:

- **Small amount of data missing:** Deletion or mean/median imputation.
- **Large amount of data missing:** Drop the feature/column or use advanced modeling.
- **Categorical data:** Mode imputation or create a new category (e.g., "Missing").
- **Time Series:** Forward fill or backward fill. [Medium +2](#)

Impact of Handling Missing Values

Handling missing values effectively is important to ensure the accuracy and reliability of our findings.

1. **Improved data quality:** A cleaner dataset with fewer missing values is more reliable for analysis and model training.
2. **Enhanced model performance:** Properly handling missing values helps models perform better by training on complete data, leading to more accurate predictions.
3. **Preservation of Data Integrity:** Imputing or removing missing values ensures consistency and accuracy in the dataset, maintaining its integrity for further analysis.
4. **Reduced bias:** Addressing missing values prevents bias in analysis, ensuring a more accurate representation of the underlying patterns in the data.

(ii) Noise smoothing

Noise smoothing techniques in data mining remove random errors, outliers, or fluctuations from datasets, enhancing pattern recognition and analysis accuracy.

Key Noise Smoothing Techniques

1. **Binning Method (Local Smoothing):** Data binning or bucketing is a data preprocessing method used to minimize the effects of small observation errors. The original data values are divided into small intervals known as bins and then they are replaced by a general value calculated for that bin. This has a smoothing effect on the input data and may also reduce the chances of overfitting in the case of small

datasets

Why Binning is Important?

1. **Data Smoothing:** Binning helps reduce the impact of minor observation variations, effectively smoothing the data.
 2. **Outlier Mitigation:** It reduces the influence of outliers by grouping values into bins.
 3. **Improved Analysis:** Discretizing continuous data simplifies data analysis and enables better visualization.
 4. **Feature Engineering:** Binned variables can be more intuitive and useful in predictive modeling.
- **Smoothing by Bin Means:** Each value in a bin is replaced by the mean value of that bin.
 - **Smoothing by Bin Medians:** Each bin value is replaced by the median value of the bin.
 - **Smoothing by Bin Boundaries:** Minimum and maximum values in a bin are identified, and values are replaced by the closest boundary.

Sorted data for price (in dollars): 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

Partition using equal frequency approach:

- Bin 1 : 4, 8, 9, 15
- Bin 2 : 21, 21, 24, 25
- Bin 3 : 26, 28, 29, 34

Smoothing by bin means:

- Bin 1: 9, 9, 9, 9
- Bin 2: 23, 23, 23, 23
- Bin 3: 29, 29, 29, 29

Smoothing by bin boundaries:

- Bin 1: 4, 4, 4, 15
- Bin 2: 21, 21, 25, 25
- Bin 3: 26, 26, 26, 34

Smoothing by bin median:

- Bin 1: 9, 9, 9, 9
- Bin 2: 23, 23, 23, 23
- Bin 3: 29, 29, 29, 29

2. **Moving Average Filter:** Replaces each data point with the average of neighboring points within a specified window, effectively reducing noise while preserving overall trends.

3. Exponential Smoothing: Assigns exponentially decreasing weights to older data points, making it highly effective for time-series data where recent data is more relevant.
4. [Gaussian Filters](#): Use weighted averages based on a Gaussian distribution, providing superior noise reduction while maintaining sharp changes (edges) in the data.
5. [Savitzky-Golay Filters](#): Use a least-squares polynomial fit over a window of points to smooth data while retaining the underlying structure

(iii) Outlier detection

(iv) Data consistency checks

♦ Techniques Used:

- Mean/Median/Mode imputation
- Regression imputation
- Binning (smoothing)
- Clustering (to detect outliers)
- Statistical methods (Z-score)

♦ Important Points:

- Improves data quality → directly impacts accuracy
- Time-consuming step in real-world systems

2. DATA INTEGRATION

✓ Purpose:

Combine data from **multiple sources**

♦ Sub-steps:

- Schema integration
- Entity identification
- Redundancy removal

♦ Techniques Used:

(i) Data fusion

Correlation analysis

- Tuple matching
- ETL

♦ Important Points:

- Handles **heterogeneous databases**
- Removes duplicate data

3. DATA SELECTION

✓ Purpose:

Select **relevant subset of data**

♦ Sub-steps:

- Attribute selection
- Record selection

♦ **Techniques Used:**

- Querying (SQL)
- Sampling
- Feature selection algorithms

♦ **Important Points:**

- Reduces computation cost
- Improves efficiency of mining

4. DATA TRANSFORMATION

✓ **Purpose:**

Convert data into suitable format for mining

♦ **Sub-steps:**

- Normalization
- Aggregation
- Generalization
- Feature construction

♦ **Techniques Used:**

- Min-Max normalization
- Z-score normalization
- Decimal scaling
- PCA (Principal Component Analysis)

♦ **Important Points:**

- Improves mining efficiency
- Reduces dimensionality

5. DATA MINING

✓ Purpose:

Extract **patterns from data**

♦ Tasks / Functionalities:

- Classification
- Clustering
- Association rule mining
- Regression
- Anomaly detection

♦ Techniques / Algorithms:

- Decision Trees (ID3, C4.5)
- Naive Bayes
- K-Means clustering
- Apriori algorithm
- SVM, Neural Networks

♦ Important Points:

- Most important step for exams
- Produces patterns, not knowledge (yet)

6. PATTERN EVALUATION

✓ Purpose:

Identify **interesting & useful patterns**

♦ Sub-steps:

- Pattern filtering

- Interestingness measurement

♦ **Techniques Used:**

- Support, Confidence, Lift
- Statistical significance testing
- Thresholding

♦ **Important Concept:**

- Interestingness measures:
 - Objective (support, confidence)
 - Subjective (user belief)

♦ **Important Points:**

- Removes redundant/unimportant patterns
- Converts patterns → meaningful insights

7. KNOWLEDGE PRESENTATION

✓ **Purpose:**

Present knowledge in **understandable form**

♦ **Sub-steps:**

- Visualization
- Reporting
- Interpretation

♦ **Techniques Used:**

- Graphs, charts, dashboards
- Data visualization tools
- OLAP cubes

♦ **Important Points:**

- Human-readable output
- Supports decision making

🔥 **KEY POINTS**

★ **1. Difference:**

KDD	Data Mining
Complete process	Single step
Includes preprocessing	Focuses on pattern extraction

★ **2. Iterative Nature**

- KDD is **not linear**
- Steps may repeat (feedback loop)

★ **3. Major Challenges**

- Data quality issues
- Scalability
- High dimensionality
- Heterogeneous data

★ **4. KDD vs ETL**

- ETL is part of preprocessing
- KDD includes ETL + Mining + Evaluation

★ **5. Real-Life Example Flow**

Bank Fraud Detection:

- Cleaning → remove missing transactions
- Integration → combine bank + card data
- Selection → choose transaction features
- Transformation → normalize data
- Mining → apply anomaly detection
- Evaluation → filter suspicious patterns
- Presentation → fraud reports

QUICK REVISION (1-LINE SUMMARY)

Step	Key Work
Cleaning	Remove noise
Integration	Combine data
Selection	Choose relevant data
Transformation	Format data
Mining	Extract patterns
Evaluation	Filter patterns
Presentation	Show knowledge

Types of Data:

- Structured, Semi-structured, Unstructured
- Spatial, Temporal, Transactional
- Data Mining Functionalities:
 - Classification, Clustering, Association, Prediction, Summarization
([Studocu](#))
- Applications:

- Market basket analysis, Fraud detection, Bioinformatics, Web mining

Data Warehouse Basics

- OLTP vs OLAP
 - Data Cube, Dimensions
 - Star Schema, Snowflake Schema
 - ETL Process (Extract, Transform, Load) ([IIT Kharagpur Facweb Portal](#))
-

UNIT 2: DATA PREPROCESSING & DATA QUALITY

Data Cleaning

- Missing values handling (mean, median, regression)
- Noise removal (binning, smoothing)

Data Integration

- Redundancy detection
- Entity resolution

Data Transformation

- Normalization:
 - Min-Max
 - Z-score
 - Decimal scaling
- Aggregation, Generalization

Data Reduction

- Dimensionality reduction
- Numerosity reduction
- Sampling

Feature Engineering

- Feature selection methods
 - Feature extraction (PCA)
-

UNIT 3: CLASSIFICATION & PREDICTION

Basic Concepts

- Supervised Learning
- Training vs Testing vs Validation
- Overfitting & Underfitting
- Bias-Variance Tradeoff ([Google Sites](#))

Classification Algorithms (VERY IMPORTANT)

- Decision Trees:
 - ID3, C4.5, CART
- Bayesian Methods:
 - Naive Bayes
- k-Nearest Neighbor (k-NN)
- Support Vector Machines (SVM)
- Artificial Neural Networks
- Logistic Regression

Ensemble Methods

- Bagging
- Boosting (AdaBoost)
- Random Forest

Evaluation Metrics

- Confusion Matrix
 - Accuracy, Precision, Recall, F1-score
 - ROC Curve, AUC ([Google Sites](#))
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UNIT 4: CLUSTERING (UNSUPERVISED LEARNING)

Concepts

- Cluster types (well-separated, density-based, etc.)
- Similarity & distance measures:
 - Euclidean, Manhattan, Cosine

Clustering Algorithms

- Partitioning:
 - K-Means, K-Medoids
- Hierarchical:
 - Agglomerative, Divisive
- Density-Based:
 - DBSCAN
- Grid-Based Methods

Cluster Evaluation

- Silhouette coefficient
 - SSE (Sum of Squared Errors)
-

UNIT 5: ASSOCIATION RULE MINING

Basic Concepts

- Frequent Itemsets
- Support, Confidence, Lift

Algorithms (VERY IMPORTANT)

- Apriori Algorithm
- FP-Growth Algorithm
- ECLAT

Advanced Topics

- Multilevel association rules
 - Constraint-based mining
 - Sequential pattern mining ([IoE SantoshB](#))
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UNIT 6: OUTLIER / ANOMALY DETECTION

- Types of outliers:
 - Global, Contextual, Collective
- Methods:
 - Statistical methods
 - Distance-based methods
 - Density-based methods

Applications:

- Fraud detection
 - Intrusion detection ([IoE SantoshB](#))
-

UNIT 7: DIMENSIONALITY REDUCTION

Techniques

- Principal Component Analysis (PCA)
- Linear Discriminant Analysis (LDA)
- Feature selection methods

Concepts

- Curse of Dimensionality ([Google Sites](#))
-

UNIT 8: ADVANCED DATA MINING TOPICS

Time-Series & Sequential Mining

- Time series analysis
- Trend & seasonality
- Sequence mining:
 - GSP, PrefixSpan ([Scribd](#))

Spatial Data Mining

- Spatial clustering
- GIS data concepts

Text Mining

- NLP basics
- TF-IDF, Bag of Words

Web Mining

- Web content mining
 - Web usage mining
 - PageRank Algorithm ([Google Sites](#))
-

UNIT 9: MACHINE LEARNING FOUNDATIONS

(IMPORTANT FOR PSUs)

- Types of learning:
 - Supervised, Unsupervised, Reinforcement
 - Model selection & validation
 - Cross-validation (K-fold)
 - Loss functions
-

UNIT 10: STATISTICAL FOUNDATIONS

- Probability basics
 - Bayes Theorem
 - Mean, Variance, Covariance
 - Gaussian distribution ([Google Sites](#))
 - Hypothesis testing
-

UNIT 11: DATA MINING TOOLS & PRACTICALS

- Tools:
 - WEKA
 - RapidMiner
 - Python (scikit-learn)
 - Data formats:
 - CSV, ARFF
 - Basic workflow:
 - Data loading → preprocessing → modeling → evaluation
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UNIT 12: IMPORTANT THEORY QUESTIONS (PSU FOCUS)

Prepare these well:

- Difference: Classification vs Clustering
- Apriori vs FP-Growth
- OLTP vs OLAP
- Overfitting vs Underfitting
- Bias vs Variance
- Supervised vs Unsupervised learning
- Data Mining vs Data Warehousing